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RESEARCH ARTICLE

A Cross-Layer FL-Based Clustering Protocol to Support Multicast Routing in IoT-Enabled MANETs With CF-mMIMO

AMALIA AMALIA[1], YUSHINTIA PRAMITARINI[1], (Graduate Student Member, IEEE), RIDHO HENDRA YOGA PERDANA[1], (Graduate Student Member, IEEE), KYUSUNG SHIM[2], (Member, IEEE), AND BEONGKU AN[3], (Senior Member, IEEE)

[1]Department of Software and Communications Engineering, Graduate School, Hongik University, Sejong 30016, Republic of Korea
[2]School of Computer Engineering and Applied Mathematics, Hankyong National University, Anseong, Gyeonggi 17579, Republic of Korea
[3]Department of Software and Communications Engineering, Hongik University, Sejong 30016, Republic of Korea

Corresponding author: Beongku An (beongku@hongik.ac.kr)

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ABSTRACT This paper proposes a novel cross-layer federated learning (FL)-based clustering (CFLC) protocol to support multicast routing in internet of things (IoT)-enabled mobile ad hoc networks (MANETs) with cell-free massive multiple-input multiple-output (CF-mMIMO). The proposed CFLC protocol leverages cross-layer and FL approaches to enhance network stability and connectivity by optimizing cluster head (CH) selection and cluster formation. The cross-layer design integrates physical layer information such as mobility (speed and direction), position, channel capacity, and remaining energy, with network layer information (connectivity) to maximize the cost function value for cluster formation. We design the FL model to improve the clustering performance and satisfy future mobile network requirements. Specifically, during the CH selection step, FL can decide which nodes should be elected as CHs or cluster members (CMs) by using classification. In the cluster formation step, FL addresses a regression problem by optimizing the cost function weights for parameters such as mobility similarity, link quality, remaining energy, and channel capacity to decide which CH each node should follow. The simulation results show that the proposed CFLC protocol outperforms the benchmark protocols in terms of connectivity, scalability, and control overhead. Additionally, the results indicate that the CFLC protocol performs particularly well when using the reference point group mobility (RPGM) model, highlighting its advantage over the random waypoint (RWP) mobility model in maintaining network stability and connectivity.

INDEX TERMS Cell-free massive MIMO, clustering, cross-layer, federated learning, IoT-enabled mobile ad hoc network.

I. INTRODUCTION

The 5G networks are designed to meet the demands of modern applications, offering capabilities such as high bandwidth, ultra-reliable low latency communication (URLLC), and massive device connectivity [1]. These features make 5G an ideal foundation for supporting the rapid growth of the internet of things (IoT), particularly in highly dynamic environments where stable, real-time communication is needed. In such environments, mobile ad-hoc networks (MANETs) have emerged as a solution, offering decentralized and dynamic connectivity without relying on the fixed infrastructure for IoT applications [2]. This capability is particularly beneficial in environments where devices constantly move and network conditions change frequently.

Despite these benefits, IoT-enabled MANETs face significant challenges in maintaining stable connections due to the dynamic and random movement of each node and

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the absence of centralized nodes to manage the network architecture [3]. Advanced wireless communication technologies like cell-free massive multiple-input multiple-output (CF-mMIMO) have been introduced to address these challenges [4], [5]. CF-mMIMO enhances network performance by supporting high-density and high-mobility scenarios, eliminating the interference limitations associated with cell boundaries. However, CF-mMIMO introduces a new issue like the pairing problem, where nodes must be efficiently paired with access points to maintain optimal network performance and minimize interference [4]. In highly dynamic MANETs, where nodes frequently change their position, this pairing must adapt rapidly to ensure stable communication.

Clustering offers a promising solution by organizing nodes into manageable groups, in which each group is managed by a cluster head (CH) [6]. This approach simplifies the pairing process between access points and users. Additionally, clustering provides a robust underlying structure to support multicast routing that is essential for efficient data transmission from a single source to multiple destinations. By grouping nodes into clusters, this network can manage each node more effectively which can enhance multicast route connectivity, reduce control overhead, and allow the CH of each cluster to handle its cluster member (CM) nodes [7]. Gateway (GW) nodes also allow inter-cluster communication that can make the connectivity robust even if the CHs are out of range [8], [9]. Despite its advantages, clustering faces several challenges, including optimal CH election and cluster formation. In order to improve the performance of these issues, the technology evolution brings cross-layer design [10]. However, the cross-layer design uses a lot of network information, which leads the computational complexity and processing time issues. Thus, an effective CH selection and cluster formation process is required to optimize energy utilization.

To address these challenges, more adaptive methods with lower computational complexity and reduced processing time are needed. Machine learning, particularly a data-driven approach, has emerged as a powerful tool in this context [11]. Unlike traditional clustering algorithms that depend on extensive iterations for optimization [12], advanced machine learning techniques, such as federated learning (FL), provide a promising solution for optimizing clustering processes in IoT-enabled MANETs with CF-mMIMO. By leveraging a distributed framework, FL utilizes deep neural networks (DNNs) to analyze complex data patterns across diverse sources within the network, facilitating real-time decision-making [13]. This approach significantly improves optimization speed, thereby reducing execution time, which is a primary focus of this paper. FL also enables decentralized processing at local nodes, reducing delays associated with centralized data aggregation while preserving data privacy by processing data locally [14], [15]. Moreover, combining FL with a cross-layer approach provides richer information by leveraging information from multiple layers. This integration makes the clustering process more efficient and adaptable to network conditions, thereby enhancing overall stability and connectivity.

A. RELATED WORKS AND MOTIVATIONS

Clustering in ad-hoc networks has been extensively studied to achieve strong stability and connectivity while minimizing control overhead and delay to ensure network scalability and efficient routing. One of the well-known clustering algorithms is the lowest ID clustering algorithm [16], in which nodes with the lowest IDs are chosen as CHs. Another approach is the highest connectivity clustering algorithm proposed by [17] that is designed to select cluster heads based on the highest number of neighbors (i.e., the highest degree of connectivity) of nodes rather than their IDs. Both the lowest ID and highest connectivity clustering algorithms are simpler but do not adapt well to the changes in network topology. The authors of [18] proposed the weighted clustering algorithm (WCA), which considers multiple parameters such as node connectivity, mobility, and energy levels for CH selection. However, this method has limitations, particularly its susceptibility to frequent cluster re-configurations due to high node mobility, leading to increased overhead.

An energy and trust management framework for MANETs utilizing a k-means clustering algorithm was proposed by [19] to save energy and manage trust by detecting and replacing unstable cluster heads through a self-configurable mechanism. However, due to its static nature, the framework's reliance on k-means clustering might not adapt well to highly dynamic network conditions. The authors of [20] proposed a graph kernel based clustering algorithm (GKCA) that uses graph theory to improve network performance and scalability, but it assumes that graph models can effectively capture node mobility patterns, which is not always feasible in dynamic environments. In contrast, the author of [21] proposed dynamic adaptability and efficient cluster management by using a centralized control mechanism that forms clusters on demand. It reduces the number of unnecessary control message exchanges to optimize the utilization of resources. However, a central node in this design introduces a single point of failure that disrupts the network in case the central node goes down.

The above approaches have been effective to some extent, but recent studies have introduced advanced techniques like bio-inspired algorithms for optimized cluster formation. For instance, [22] used gray wolf optimization to enhance clustering stability while [23] used particle swarm optimization (PSO) to improve cluster stability and extend network lifetime. However, these algorithms typically require significant iterations to reach the optimal cost function, which can be time-consuming. Contrary to this, machine learning techniques offer a more efficient solution, especially for large-scale, high-dimensional problems. The authors of [24] proposed a deep learning-based dynamic stable CH selection that leverages long short-term memory (LSTM) networks to

predict and select the most stable CHs based on historical mobility patterns and network conditions. The proposed method not only improves the stability and reliability of the network but also optimizes energy consumption, leading to a prolonged network lifetime. However, their centralized approach may not fully utilize the distributed nature of ad-hoc networks. Similarly, [25] proposed a decentralized hierarchical federated learning (DHFL) framework designed to optimize CH selection and clustering, reducing training time, but it did not consider multiple parameters for cluster formation.

The above-mentioned works have explored various clustering techniques and machine-learning applications to address stability and connectivity challenges. However, many of these approaches involve extensive iterations during the optimization process, leading to significant delays that can disrupt URLLC services. Additionally, limited attention has been given to integrating FL with clustering methods in CF-mMIMO environments. Different from other works [26], [27] that implement CF-mMIMO in a static network environment, our approach goes a step further by considering the dynamic and often unpredictable nature of MANETs. Motivated by the need to minimize delay and maintain URLLC standards, three fundamental questions arise, which will be addressed in this paper: (1) How can network stability and connectivity be enhanced while reducing control overhead in a high-mobility MANET environment with frequent topology changes? (2) How can cross-layer information be effectively leveraged to enhance network stability and connectivity by optimizing CH selection and cluster formation in a MANET environment? (3) How to utilize a data-driven approach, such as FL, instead of an iteration-based approach to improve clustering performance?

B. CONTRIBUTIONS AND ORGANIZATION

We can see that there is a gap in the literature regarding efficient clustering and maintaining network stability in IoT-enabled MANETs with CF-mMIMO, particularly under dynamic and high-mobility conditions. Traditional clustering approaches often require extensive iterations for optimization, leading to increased execution time and delays that disrupt URLLC services. To bridge this gap, we propose a cross-layer federated learning-based clustering (CFLC) protocol that leverages cross-layer and FL to optimize CH selection and cluster formation. This approach aims to enhance network stability and connectivity while reducing control overhead, aligning with the stringent performance requirements of 5G and beyond 5G wireless networks. The main contributions of this paper can be summarized as follows:

- We propose a novel CFLC protocol to support multicast routing in IoT-enabled MANETs with CF-mMIMO. This protocol leverages cross-layer and FL to enhance network stability and connectivity by optimizing CH selection and cluster formation.

TABLE 1. List of abbreviations.

Abbreviations	Definition
AP	Access points
CFLC	Cross-layer federated learning-based clustering
CF-mMIMO	Cell-free massive multiple-input multiple-output
CH	Cluster head
CHI	Cluster head information
CM	Cluster member
CPU	Central processing unit
DNN	Deep neural network
ELU	Exponential linear unit
FL	Federated learning
GPS	Global positioning system
GW	Gateway
HC	Highest connectivity
INFO	Information
IoT	Internet of things
JOIN	Join cluster
LI	Lowest ID
MAC	Medium access control
MANETs	Mobile ad hoc networks
MSE	Mean square error
RMSE	Root mean square error
RPGM	Reference point group mobility
RSSI	Received signal strength indicator
RWP	Random waypoint
UE	User equipment
URLLC	Ultra-reliable low-latency communications

- To enhance the network performance, we propose a cross-layer design that integrates physical layer information such as mobility (speed and direction), position, channel capacity, and remaining energy with network layer information (connectivity) that maximizes the cost function value for cluster formation. Additionally, we consider the selection of GW nodes to ensure seamless communication continuity between clusters, especially to support multicast routing when communication extends beyond the cluster range.
- We design the FL model to improve the clustering performance and satisfy the future mobile network requirement by utilizing mobile edge computing access points (APs) for processing local models and a central processing unit (CPU) for model aggregation. Specifically, during the CH selection step, FL can decide which nodes should be elected as CHs or CMs by using classification. In the cluster formation step, FL addresses a regression problem by optimizing the cost function weights for parameters such as mobility similarity, link quality, remaining energy, and channel capacity to decide which CH each node should follow.
- The simulation results show that the proposed CFLC protocol outperforms the benchmark protocols in terms of connectivity, scalability, and control overhead, demonstrating improved clustering performance. Additionally, the results indicate that the CFLC protocol performs particularly well when using the reference point group mobility (RPGM) model, highlighting its advantage over the random waypoint (RWP) mobility model in maintaining network stability and connectivity.

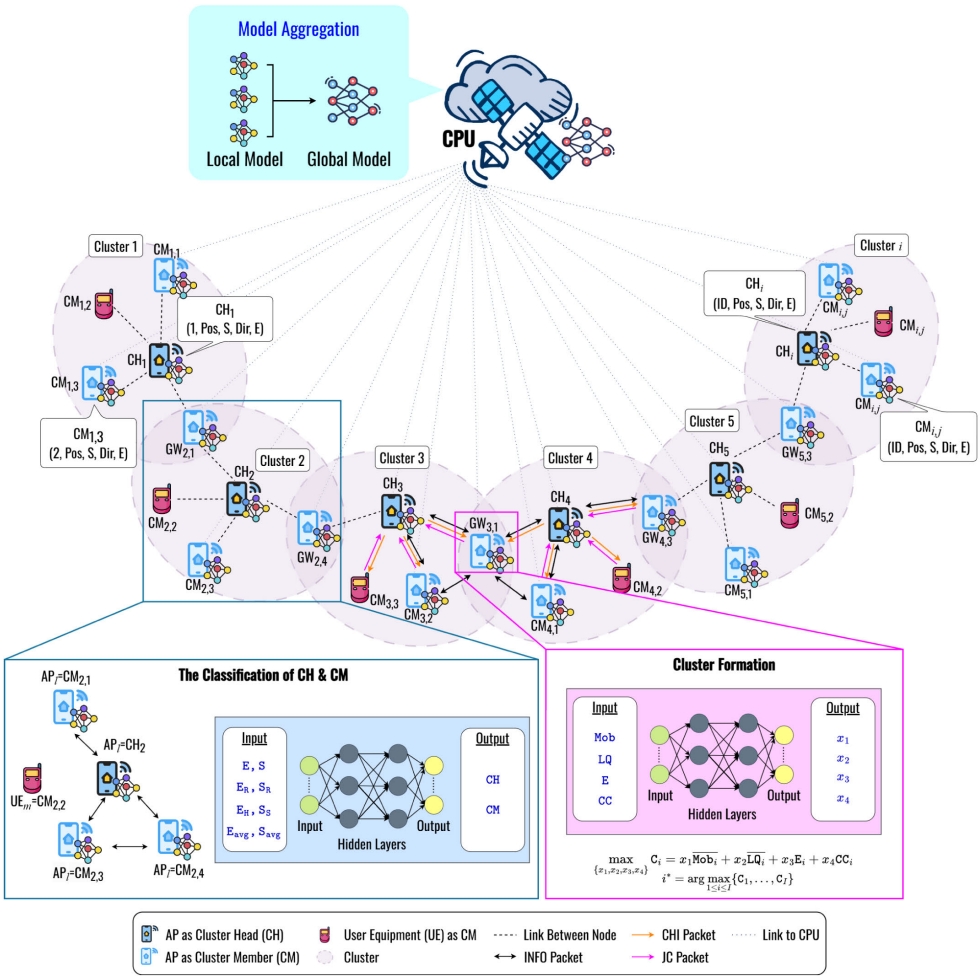


FIGURE 1. The basic concepts of the proposed cross-layer federated learning-based clustering (CFLC) protocol.

The rest of the paper is organized as follows: Section II explains the proposed system model that consists of the cross-layer design, the classification of CH and CM, cluster formation, and the proposed CFLC protocol. Section III explains the proposed FL. Section IV discusses the performance evaluation that consists of simulation environments and network parameters, performance metrics, and numerical results, and Section V concludes the paper. In addition, for clarity and ease of reference, the primary abbreviations used in this paper are listed in Table 1.

II. SYSTEM MODEL

In this section, we introduce a novel cross-layer FL-based clustering protocol as shown in Fig. 1 to support multicast routing in IoT-enabled MANETs with CF-mMIMO, aiming to enhance network stability and connectivity. This network includes two different types of nodes, $L = \{\mathrm{AP}_l\}_{l=1}^{L}$ represented by a set of mobile edge computing access points node AP and $M = \{\mathrm{UE}_m\}_{m=1}^{M}$ represented by a set of user equipment node UE. These nodes are organized into clusters, $Cl_i \triangleq \{i = 1, 2, \ldots, I\}$ and each cluster has UE and multiple AP as cluster members. Each node has information such as nodes identifier (ID), position (Pos), speed (S), direction (Dir), and remaining energy (E). This is based on the assumption that each node can know the position information using a global positioning system (GPS). In the practical implementation of the proposed protocol, we assume that only APs are eligible to serve as CHs. This ensures that CH selection is limited to nodes with greater stability and resources. However, despite this careful selection of CHs, the dynamic mobility of nodes still leads to frequent changes in network topology, presenting significant challenges in maintaining stable clusters.

To effectively address these challenges, the proposed CFLC protocol enhances network stability and connectivity by forming clusters based on node mobility using cross-layer design. This process begins with classifying nodes as either CHs or CMs before initiating cluster formation. Using FL for classification allows for proactive and informed decision-making, ensuring that nodes with optimal characteristics, such as high remaining energy and stable speed, are chosen

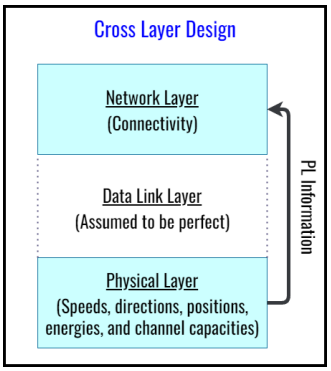


FIGURE 2. A detailed depiction of the cross-layer design for the proposed CFLC protocol.

as CHs. Once nodes are classified into CHs and CMs, the proposed CFLC protocol proceeds with the cluster formation process by optimizing a cost function that addresses the regression problem. The regression model optimizes key parameters such as mobility similarity, link quality, remaining energy, and channel capacity to select the most suitable CH to follow. This dual approach of classification and regression not only enhances the stability and performance of the network but also minimizes the need for frequent re-clustering.

A. CROSS LAYER DESIGN

In our cross-layer design, we leverage information from the physical layer and network layer while assuming an ideal data link layer as shown in Fig. 2. This strategy enables our proposed CFLC protocol to establish and maintain clusters efficiently, enhancing the overall performance. By conveying information from the physical layer to the network layer, our approach integrates a comprehensive set of parameters to improve the accuracy and efficiency of the clustering process. Our cross-layer design is based on the assumption that each node possesses accurate position information obtained through the GPS. The roles of the physical and network layers in detail can be explained as follows:

- **Physical Layer:** Provides critical information on node's mobility (speeds and directions), positions, remaining energies, and channel capacities. This information is crucial for understanding node behavior and assessing its capability.
- **Network Layer:** Utilizes the information obtained from the physical layer, including mobility parameters (speed, direction), and node positions to calculate mobility similarity and link quality. Thus, information such as mobility similarity, link quality, remaining energy, and channel capacity is utilized for cost function optimization, enabling the selection of the most optimal CH to follow for the cluster formation process.

B. THE PROPOSED CFLC PROTOCOL

In this subsection, we explain the details of the proposed CFLC protocol. The proposed CFLC protocol considers a

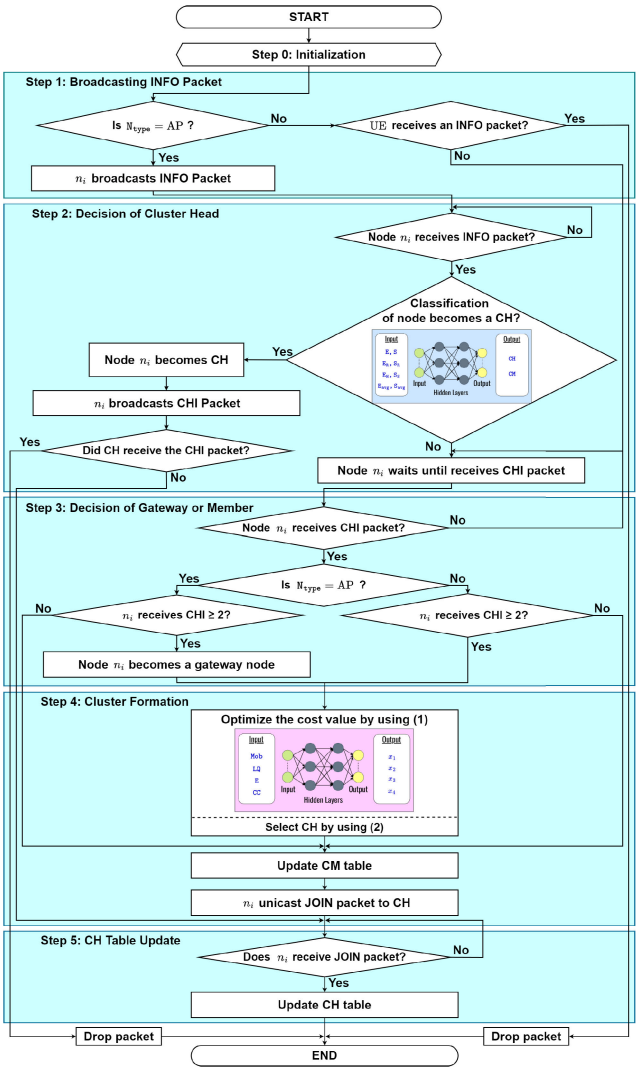


FIGURE 3. The detailed flowchart of the proposed CFLC protocol, illustrates the step-by-step process of CH selection and cluster formation.

cross-layer design that uses both physical layer information and network layer information to form a cluster. In addition, we consider the classification of CH and CM based on node information such as remaining energy and speed. Fig. 3 shows the flowchart of the proposed CFLC protocol which can be explained as follows:

- **Step 0: Initialization**
  - Each node obtains mobility (speed and direction), position, channel capacity, and remaining energy from the physical layer. This information is used for the clustering process at the network layer.
  - In addition, all APs independently train their local models until convergence. Once the local models are trained, they send these models to the CPU for FL aggregation. The CPU aggregates the local models to form global models and then disseminates this to all APs.
  - Furthermore, the APs disseminate the global model to the nearest UEs within their transmission area. This

ensures that all nodes have the necessary information and models before proceeding to the clustering process and proceed to **Step 1**.

- **Step 1: Broadcasting INFO packet**
  Each AP node n_i periodically disseminates node information such as speed and remaining energy. Subsequently, n_i generates and broadcasts node information (INFO) packets to its neighboring nodes while simultaneously receiving the same packets. The following fields are included in the INFO packet:

$$\langle \mathtt{Type}, \mathtt{S_{ID}}, \mathtt{D_{ID}}, \mathtt{E}, \mathtt{S}, \mathtt{N_{type}} \rangle$$

  where $\mathtt{Type}$ represents packet type, $\mathtt{S_{ID}}$ represents source node ID, $\mathtt{D_{ID}}$ represents destination node ID, $\mathtt{E}$ represents the remaining energy of the node, $\mathtt{S}$ represents the node speed, and $\mathtt{N_{type}}$ represents the type of node (AP or UE), respectively. Then, proceeds to **Step 2**.
  - If the UE node receives an INFO packet, it drops the packet. Otherwise, it waits until it receives the cluster head information (CHI) packet in **Step 3**.
- **Step 2: Decision of Cluster Head**
  After receiving the INFO packet from the neighborhood, the FL model classifies each node as either a CH or CM. The detailed classification process is elaborated in Section III-A1.
  - If the node classification results designate n_i as the CH, it broadcasts the CHI packet to its neighboring nodes. Otherwise, n_i becomes a CM and waits until it receives a CHI packet. The CHI packet contains the following fields:

$$\langle \mathtt{Type}, \mathtt{S_{ID}}, \mathtt{D_{ID}}, \mathtt{Pos}, \mathtt{E}, \mathtt{S}, \mathtt{Dir}, \mathtt{N_{type}} \rangle$$

    where $\mathtt{Type}$ represents packet type, $\mathtt{S_{ID}}$ represents source node ID, $\mathtt{D_{ID}}$ represents destination node ID, $\mathtt{Pos}$ represents the node position, $\mathtt{E}$ represents the remaining energy of the node, $\mathtt{S}$ represents the node speed, $\mathtt{Dir}$ represents the node direction, and $\mathtt{N_{type}}$ represents the type of node (AP or UE), respectively. Then, proceed to **Step 3**.
  - If CH receives the CHI packet, CH will drop it. Otherwise, CH waits until it receives a JOIN packet in **Step 4**.
- **Step 3: Decision of Gateway or Member**
  Node n_i can receive multiple CHI packets, leading to the following process:
  - If node n_i is identified as an AP node and receives two or more CHI packets, it becomes a GW node and determines which CH to follow in **Step 4**. Otherwise, it directly becomes a CM.
  - If node n_i is identified as a UE node and receives two or more CHI packets, it becomes a CM by determining which CH to follow in **Step 4**. Otherwise, it directly becomes a CM.
- **Step 4: Cluster Formation**
  When node n_i (AP or UE) receives two or more CHI packets, it will determine which CH to follow by selecting one with the highest cost function value. The proposed cost function can be expressed as

$$\max_{\{x_1,x_2,x_3,x_4\}} \mathtt{C}_i = x_1\overline{\mathtt{Mob}_i} + x_2\overline{\mathtt{LQ}_i} + x_3\mathtt{E}_i + x_4\mathtt{CC}_i, \tag{1a}$$

$$\text{s.t} \quad x_1 + x_2 + x_3 + x_4 = 1, \tag{1b}$$

$$x_1 > 0, x_2 > 0, x_3 > 0, x_4 > 0, \tag{1c}$$

$$\overline{\mathtt{Mob}_i} \geq \overline{\mathtt{Mob}_{th}}, \tag{1d}$$

$$\overline{\mathtt{LQ}_i} \geq \overline{\mathtt{LQ}_{th}}, \tag{1e}$$

$$\mathtt{E}_i \geq \mathtt{E}_{th}, \tag{1f}$$

$$\mathtt{CC}_i \geq \mathtt{CC}_{th}, \tag{1g}$$

  where $\overline{\mathtt{Mob}_i}$, $\overline{\mathtt{LQ}_i}$, $\mathtt{E}_i$, and $\mathtt{CC}_i$ are average mobility similarity, average link quality, remaining energy, and channel capacity, respectively. Equation (1b) and (1c) establish the requirement that the total weight must equal one and have a positive number. Additionally, (1d) specifies that mobility similarity should be greater or equal to the threshold, (1e) specifies that the link lifetime should be greater or equal to the threshold while (1f) specifies the condition that the remaining energy must be greater or equal to the threshold. Finally, (1g) outlines that the channel capacity must be greater or equal to the threshold. To determine the optimal weight in (1b), which typically requires several iterations, we utilize FL with the DNN model described in Section III-A2. This approach can deliver faster and more accurate results compared to conventional iterative methods when determining optimal values. Then, the selection of the optimal CH can be mathematically formulated as

$$i^* = \arg\max_{1 \leq i \leq I} \{\mathtt{C}_1, \ldots, \mathtt{C}_I\} \tag{2}$$

  where $\mathtt{C}_i$ represents the cost function value for cluster head i. The optimal cluster head i^* is the one that maximizes the cost function among all potential cluster heads 1 to I. Consequently, n_i will update the CM table as shown in Table 2, where $\mathtt{CM_{ID}}$ represents the cluster member identifier, $\mathtt{CH_{ID}}$ represents the identifier of the cluster head that the cluster member belongs to, and $\mathtt{N_{type}}$ represents the type of nodes (AP or UE), respectively. Furthermore, a JOIN packet is sent to the chosen CH, indicating its decision to follow that CH. The following fields are included in the JOIN packet:

$$\langle \mathtt{Type}, \mathtt{S_{ID}}, \mathtt{D_{ID}}, \mathtt{N_{type}} \rangle$$

  where $\mathtt{Type}$ represents packet type, $\mathtt{S_{ID}}$ represents source node ID, $\mathtt{D_{ID}}$ represents destination node ID, and $\mathtt{N_{type}}$ represents the type of node, respectively. Then, proceeds to **Step 5**.

- **Step 5: CH Table Update**
  If node n_i receives a JOIN packet, it will update the CH table as shown in Table 3. In this table, $\mathtt{CH_{ID}}$ represents the cluster head identifier, $\mathtt{CM_{ID}}$ represents the identifier of the cluster member that belongs to the cluster managed by the cluster head, and $\mathtt{N_{type}}$ represents the type of node, respectively. Otherwise, the node n_i waits until it receives a JOIN packet.

TABLE 2. Cluster member table of the proposed CFLC protocol.

$\mathtt{CM_{ID}}$	$\mathtt{CH_{ID}}$	$\mathtt{N_{type}}$

TABLE 3. Cluster head table of the proposed CFLC protocol.

$\mathtt{CH_{ID}}$	$\mathtt{CM_{ID}}$	$\mathtt{N_{type}}$

Finally, the clustering process is completed and clusters are formed. All APs continuously update their local model to reflect the latest network state. In the case of UE nodes, they update their local models and send them to their associated CH APs. Periodically, these updated local models are sent to the CPU for aggregation. In addition, data packets can be sent from a source node to a destination node by initiating the routing process using the routing protocol algorithm. Our future works will focus on the routing algorithm, specifically determining the optimal data transmission route. The details of the packet involved in the proposed CFLC protocol are summarized in Table 4.

TABLE 4. Packet List for the proposed CFLC Protocol.

Packet Name	Field Information
INFO	$\mathtt{Type}, \mathtt{S_{ID}}, \mathtt{D_{ID}}, \mathtt{E}, \mathtt{S}, \mathtt{N_{type}}$
CHI	$\mathtt{Type}, \mathtt{S_{ID}}, \mathtt{D_{ID}}, \mathtt{Pos}, \mathtt{E}, \mathtt{S}, \mathtt{Dir}, \mathtt{N_{type}}$
JOIN	$\mathtt{Type}, \mathtt{S_{ID}}, \mathtt{D_{ID}}, \mathtt{N_{type}}$

C. ENERGY CONSUMPTION MODEL

In this subsection, we consider an energy model for the proposed CFLC protocol. The energy consumed to transmit a packet through the link (i, j) based on the packet type (e.g., INFO, CHI, and JOIN packets) can be expressed as [28]

$$\mathtt{E}^t_{\mathtt{tx}} = \left(\mathtt{P_{pt}} + \frac{\mathtt{P}^{(i,j)}_{\mathtt{tx}}}{\kappa}\right)\frac{x^t}{r}, \tag{3}$$

where $t \in \{\text{INFO, CHI, JOIN}\}$ represents packet types, $\mathtt{P_{pt}}$ represents the processing circuit power for transmitting, $\mathtt{P}^{(i,j)}_{\mathtt{tx}}$ represents the transmission power required to send data from node i to the node j, κ represents the efficiency of the power amplifier, r represents data rate, and x^t represents the packet size specific to each packet type, respectively. In addition, the energy consumed to receive a packet based on the packet type is formulated as

$$\mathtt{E}^t_{\mathtt{rx}} = \frac{\mathtt{P_{pr}}}{r}x^t, \tag{4}$$

where $\mathtt{P_{pr}}$ is the power required to run the processing circuit of the receiver. Thus, the average energy consumption can be expressed as

$$\mathtt{E}_{\text{average}} = \frac{1}{NS}\sum_{s=1}^{S}\left[(\Delta^t_{s,\mathtt{tx}}\,\mathtt{E}^t_{\mathtt{tx}}) + (\Delta^t_{s,\mathtt{rx}}\,\mathtt{E}^t_{\mathtt{rx}})\right], \tag{5}$$

where N represents the total number of nodes in the network, S represents the total number of sessions, $t \in \{\text{INFO, CHI, JOIN}\}$ represents packet types, $\Delta^t_{s,\mathtt{tx}}$ represents the total number of packets of type t transmitted in the network during session s, $\Delta^t_{s,\mathtt{rx}}$ represents the total number of packets of type t received in the network during session s, $\mathtt{E}^t_{\mathtt{tx}}$ and $\mathtt{E}^t_{\mathtt{rx}}$ represent the energy required for transmission and reception of packet type t, respectively.

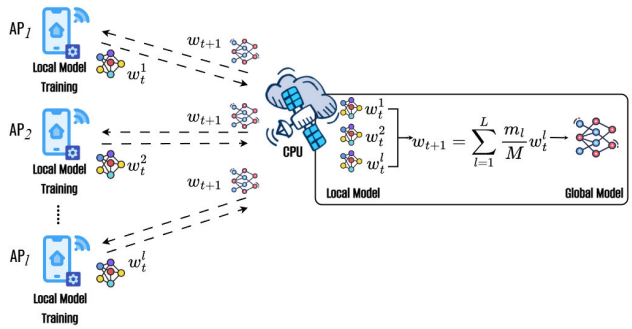


FIGURE 4. The architecture of federated learning for model aggregation in the proposed CFLC protocol.

III. FEDERATED LEARNING FOR THE PROPOSED CFLC PROTOCOL

In this section, we introduce the application of FL to machine learning for model aggregation and its complexity analysis. As shown in Fig. 4, each AP node independently trains its local model using local data. This decentralized approach ensures data privacy as raw data never leaves the node. Periodically, the AP nodes complete local training and then send their updated model parameters (such as weights and biases in a neural network) to the CPU for aggregation. During this process, the CPU combines the parameters of the local models into a single global model that encapsulates the collective knowledge from all AP nodes. This aggregated global model is then redistributed to all APs, ensuring each node benefits from the collective learning experience. By doing so, each AP node can benefit from the training data and insights gathered by all other nodes in the network. This way, even though each node trains on its local data, the collective learning process allows every node to improve its model with the broader dataset represented by the entire network. This iterative process of local training, parameter sharing, model aggregation, and redistribution of the global model repeats periodically. It ensures that the

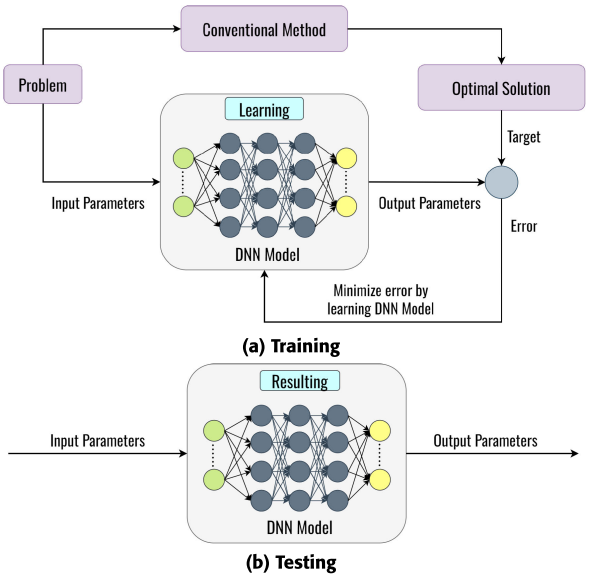


FIGURE 5. The federated learning framework incorporates a DNN model.

model continuously improves and adapts to new data while maintaining data privacy. The model aggregation to get a global model can be expressed as [29]

$$w_{t+1} = \sum_{l=1}^{L} \frac{m_l}{M} w_t^l, \tag{6}$$

where w_{t+1} represents the global model parameter after aggregation, L represents the number of participating nodes, m_l represents the number of samples held by node l, M represents the total number of samples of all nodes, and w_t^l represents the local model parameter of node l at time step t, respectively. The aggregate weights represent a global model informed by the insight from all participating nodes.

A. FEDERATED LEARNING FRAMEWORK USING DNN

FL is a widely used distributed framework that protects the privacy of each node by sharing model parameters instead of raw data. This approach enhances data confidentiality, as only the models are shared, ensuring that sensitive information remains secure on each node's device. While any machine learning method capable of learning input-output relationships can be employed, this study specifically utilizes a DNN. DNN is chosen for its robust capability to model complex and non-linear relationships between inputs and outputs, making it highly effective for tasks requiring high prediction accuracy. Fig. 5 shows the proposed FL framework based on the DNN model to obtain the optimal solution. More specifically, the proposed FL framework is used for solutions to these problems as follows:

- **Problem I:** The classification of a node becomes CH or CM.
- **Problem II:** Finding the optimal cost value for cluster formation.

We consider the general FL framework, which consists of two phases: training and testing. This framework supports continuous learning, allowing the local model that uses DNN to be periodically retrained and updated with new data during each session. It can be adapted for various tasks such as classification and regression while preserving the same overall structure. As illustrated in Fig. 5(a), the DNN model learns from the generated dataset in the training phase. This dataset is created using the conventional method, which guarantees convergence to the global optimum. The solutions obtained from this conventional method serve as the target parameters for the DNN model. The DNN is trained to learn the relationship between the input trainable parameters and these target parameters. During the training phase, the DNN's output is compared to the target values derived from the conventional method to calculate the error. This error is then minimized by updating the weights and biases of the neurons through back-propagation, iterating until the error is sufficiently minimized. After training, the DNN model can predict optimal values for new input data as depicted in Fig. 5(b). This process allows the DNN to learn and predict optimal values effectively. Notably, this framework is adaptable for classification and regression problems while maintaining the same overall structure. Further details will be provided in the following sub-subsection."

1) DNN FOR THE CLASSIFICATION OF CH AND CM

The decision for a node to become a CH or CM is based on statistical parameters, such as remaining energy and speed, extracted from INFO packets. Upon receiving these packets from its neighbors, the node aggregates the data and uses it as input for the DNN model. The input parameters for the DNN model include:

- **Node's remaining energy** ($\mathtt{E}$)**:** indicates the current energy level of the node, crucial for determining its ability to handle the additional responsibilities of a CH.
- **Speed** ($\mathtt{S}$)**:** indicates the node's mobility, which is crucial for maintaining stable clusters. Nodes with lower speeds are generally preferred as CHs to ensure cluster stability.
- **Rank of remaining energy from neighborhood** ($\mathtt{E_R}$)**:** indicates the node's remaining energy compared to its neighbors, providing a relative measure of energy efficiency.
- **Rank of speed from neighborhood** ($\mathtt{S_R}$)**:** indicates the node's speed relative to its neighbors, helping to identify the most stable nodes.
- **Highest remaining energy from neighborhood** ($\mathtt{E_H}$)**:** indicates the highest energy level among the neighboring nodes, offering a benchmark for energy comparison.
- **Slowest speed from neighborhood** ($\mathtt{S_S}$)**:** indicates the slowest speed among the neighboring nodes, identifying the most stable node in terms of mobility.
- **Average remaining energy from neighborhood** ($\mathtt{E_{avg}}$)**:** indicates the average energy level of the

neighboring nodes, offering an insight into the general energy availability within the network.

- **Average speed from neighborhood** ($\mathtt{S}_{\mathtt{avg}}$)**:** indicates the average speed of the neighboring nodes, reflecting the general mobility pattern in the vicinity.

The DNN model processes these input features through its deep learning layers to recognize patterns and relationships that indicate whether a node should be a CH or a CM. We first define a feature vector for node i, which encapsulates the relevant input parameters to represent the classification process. The feature vector for a node i can be defined as

$$\mathtt{x}_i = \left[\mathtt{E}_i, \mathtt{S}_i, \mathtt{E}_{\mathtt{R}i}, \mathtt{S}_{\mathtt{R}i}, \mathtt{E}_{\mathtt{H}i}, \mathtt{S}_{\mathtt{S}i}, \mathtt{E}_{\mathtt{avg}i}, \mathtt{S}_{\mathtt{avg}i}\right]. \tag{7}$$

This feature vector $\mathtt{x}_i$ represents the input data used to classify node i as either a CH or a CM. The DNN model then applies a linear transformation to this input vector. The linear transformation can be expressed as

$$\mathtt{z}_i = \mathbf{W} \cdot \mathtt{x}_i + \mathtt{b}, \tag{8}$$

where $\mathbf{W} \in \mathbb{R}^{2\times 8}$ is a weight matrix with dimensions corresponding to the two classes (CH or CM) and the eight input features and $\mathtt{b} \in \mathbb{R}^2$ is the bias vector, respectively. The result of this linear transformation, $\mathtt{z}_i$, represents the raw scores or logits for each class for node i. To convert these scores into probabilities, the softmax function is applied as follows [30]

$$P(y_i = c \mid \mathtt{x}_i) = \frac{\exp(\mathtt{z}_{i,c})}{\sum_{j=1}^{2} \exp(\mathtt{z}_{i,j})}, \tag{9}$$

where $P(y_i = c \mid \mathtt{x}_i)$ is the probability that node i belongs to class c (either CH or CM), $\mathtt{z}_{i,c}$ is the score for class c, and the denominator is the sum of the exponentials of the scores across both classes, ensuring that the output probabilities sum to 1, respectively. The node is then classified based on the class with the highest probability as follows

$$\hat{y}_i = \arg \max_{c\in\{\mathrm{CH},\,\mathrm{CM}\}} P(y_i = c \mid \mathtt{x}_i), \tag{10}$$

where $\hat{y}_i$ is the predicted class for node i. The softmax function enables the model to make a probabilistic decision, selecting the class with the highest likelihood given the input feature vector $\mathtt{x}_i$. Thus, to address Problem I, we proposed the architecture of a fully connected DNN model to classify nodes as either CH or CM as shown in Fig. 6. The DNN model that we employ is a feed-forward neural network with 1×8 dimensional input layers, several hidden layers with numerous hidden neurons connected by symmetric weights, and 1×2 dimensional output layers.

The structure of the layers employed in the DNN model enhanced the system's performance as shown in Table 5. We used the exponential linear unit (ELU) activation function for the input and hidden layers to accelerate learning by mitigating the vanishing gradient problem, providing faster and more stable convergence [31]. Moreover, the softmax activation function is applied to the output layer to convert logits into probabilities that sum to 1, allowing the model to classify each node as a CH or CM based on the highest probability.

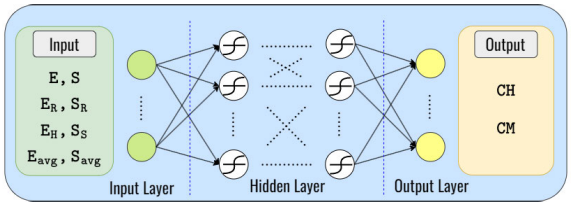


FIGURE 6. The proposed DNN model architecture for the classification of CH and CM.

TABLE 5. The layer structure of the DNN model for classification of CH and CM.

	Size	Activation Function
Input	8	ELU
Layer 1	50	ELU
Layer 2	60	ELU
Layer 3	50	ELU
Output	2	SOFTMAX

2) DNN FOR CLUSTER FORMATION

In the cluster formation process, our primary objective is to maximize the cost function to enhance stability and connectivity. This is achieved by optimizing the weight factors related to mobility similarity, link quality, remaining energy, and channel capacity. First, we consider mobility similarity to capture the similarity of the mobility features (relative velocity and relative acceleration) between two nodes within their communication range [32]. As a result, the cluster stays stable for longer periods, which reduces the requirement for frequent re-clustering. The relative velocity of nodes i and j can be defined as [33]

$$\mathtt{RV}_{(i,j)} = \left(1 - \frac{|\vec{v}_i - \vec{v}_j|}{v_{\max}}\right), \tag{11}$$

where v_{max} is the node's maximum speed. In addition, the relative acceleration of nodes i and j can be defined as

$$\mathtt{RA}_{(i,j)} = \left(1 - \frac{|\vec{a}_i - \vec{a}_j|}{a_{\max}}\right), \tag{12}$$

where a_{max} is the node's maximum acceleration. Therefore, mobility similarity between two nodes can be defined as

$$\mathtt{Mob}_{(i,j)} = \mathtt{RV}_{(i,j)} \times \mathtt{RA}_{(i,j)}. \tag{13}$$

For a node i with n neighbors, we can define the average mobility similarity $\overline{\mathtt{Mob}_i}$ as the mean mobility similarity of node i with each of its n neighbors and can be calculated as

$$\overline{\mathtt{Mob}_i} = \frac{\sum_{j=1}^{n} \mathtt{Mob}_{(i,j)}}{n}. \tag{14}$$

High average mobility similarity indicates that node i and its neighbors move similarly, suggesting stable and predictable

connections, which can lead to more stable clusters. Second, we consider link quality that can be expressed as [22]

$$\mathtt{LQ}_{(i,j)} = \frac{R + \beta \times \mathrm{dis}_{(i,j)}}{|\vec{v}_i - \vec{v}_j|}, \tag{15}$$

where R, $\mathrm{dis}_{i,j}$, and β are the transmission range, the distance between two nodes, and the sign function with the value 1 when nodes move toward each other and -1 otherwise, respectively. In addition, $\vec{v}_i$ and $\vec{v}_j$ are the velocities between two nodes. For a node i with n neighbors, we can define the average link quality $\overline{\mathtt{LQ}_i}$ as the mean link quality of node i with each of its n neighbors and can be calculated as

$$\overline{\mathtt{LQ}_i} = \frac{\sum_{j=1}^{n} \mathtt{LQ}_{(i,j)}}{n}. \tag{16}$$

Using link quality in the clustering process can enhance the stability and efficiency of the network. Nodes with high average link quality indicate a long-lasting, stable link with its neighbors. Third, we consider the remaining energy at node i-th which can be expressed as

$$\mathtt{E}_i = \mathtt{E}_{current_i} - \mathtt{E}_{mode} \quad \text{[joule]}. \tag{17}$$

where $\mathtt{E}_{current_i}$ represents the current remaining energy of the node n_i while E_{mode} represents the energy mode with mode $\in$ [0,1]. Specifically, a value of 0 indicates that the node is in transmission mode, while a value of 1 indicates that the node is in reception mode. Lastly, we consider channel capacity to maintain high communication quality, thereby reducing the likelihood of transmission errors and minimizing the need for re-transmissions. The channel capacity can be obtained as [34]

$$\mathtt{CC}_i = \log_2(1 + \mathtt{SNR}), \tag{18}$$

where SNR is received signal power at each node. The received power at each node is determined using the received signal strength indicator (RSSI), a metric measured upon node packet reception. Higher SNR values indicate better signal quality, which leads to higher channel capacity. Therefore, to address Problem II, we also utilize a DNN model to find the optimal solution for cluster formation.

The main goal of this approach is to determine the optimal cost function weight values. This method differs from the conventional optimization method for solving the problem in Eq. (1), which requires multiple iterations. The DNN model we employ is a feed-forward neural network with a size 1×4 as input parameters for the neural network. The neural network designed with several hidden layers and numerous hidden neurons connected by symmetric weight learns the complex functions necessary for this optimization. The output consists of the optimal weight values of x_1, x_2, x_3, and x_4 with the size 1×4. We designed the DNN model architecture to solve a multivariate regression problem, which is more complex than a single regression problem because it predicts multiple values simultaneously as shown in Fig. 7. This added complexity makes it a more challenging task. The linear activation function is used in the output layer because it is suitable for regression tasks that require predicting continuous numerical values. Table 6 presents the layer structure employed in this DNN model.

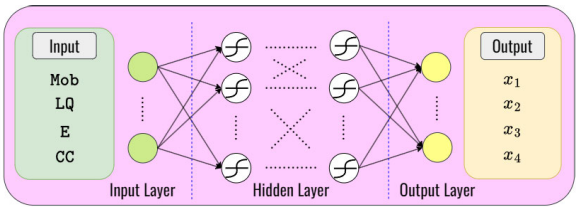


FIGURE 7. The proposed DNN model architecture for the cluster formation.

TABLE 6. The layer structure of the DNN model for cluster formation.

	Size	Activation Function
Input	4	ELU
Layer 1	40	ELU
Layer 2	100	ELU
Layer 3	70	ELU
Output	4	LINEAR

B. COMPLEXITY ANALYSIS

In this subsection, we describe the complexity analysis of the FL process in the proposed CFLC protocol. The worst-case complexity at each AP node that involves feed-forward and back-propagation processes is $\mathcal{O}(RYU^2)$, where R represents the total number of sessions to train the local model, Y represents the number of layers in the considered FL model, and U represents the number of neurons in each layer, respectively. Suppose the network consists of one CPU and participating APs denoted as L, with the total number of model parameters denoted as P. The uplink communication overhead associated with transmitting P model parameters from L APs to CPU is $\mathcal{O}(LP)$. Furthermore, the computational complexity at the CPU for aggregating the received P models from all L participating APs is also $\mathcal{O}(LP)$. Following the aggregation step, the global model consisting of the updated P model parameters is broadcast to all participating APs resulting in a downlink communication overhead of $\mathcal{O}(P)$. Thus, the total complexity for local training, uplink communication, model aggregation, and downlink communication is $\mathcal{O}(RYU^2 + 2LP + P)$. We summarize the complexities of the FL process as shown in Table 7. Since model exchange between APs and CPU is considered as communication overhead and operates independently from clustering, it does not impact the control overhead in the clustering process.

IV. PERFORMANCE EVALUATION

A. SIMULATION SETUP AND NETWORK PARAMETERS

In this section, we evaluate the performance of the proposed CFLC protocol through simulation testing. This approach allows us to identify and address any logical and/or theoretical issues directly within the simulation environment, which incurs no additional costs. Therefore, it is common to check the proposed clustering protocol through simulations

TABLE 7. Complexity analysis of the FL process.

Process	Complexity
Local Training	$\mathcal{O}(RYU^2)$
Uplink Communication	$\mathcal{O}(LP)$
Model Aggregation	$\mathcal{O}(LP)$
Downlink Communication	$\mathcal{O}(P)$
Total Complexity	$\mathcal{O}(RYU^2 + 2LP + P)$

before real-world testing. Table 8 presents the simulation environments and parameters. We use NS3 version 3.35 to simulate the proposed clustering protocol, where the total time of simulations is 110 seconds with 5 seconds for every session. In our deployment scenario, we set up two configurations within an urban area of 1000 m × 1000 m. Specifically, for the setup with N = 40 nodes, there are 25 APs and 15 UEs. In the setup with N = 80 nodes, the configuration includes 50 APs and 30 UEs. This allows us to evaluate and compare the protocol performances under varying node densities. The mobile nodes' initial positions are randomly distributed and they move at different speeds (15, 30, 45, 60, and 75 km/h). Each mobile node is equipped with an omnidirectional antenna, and its maximum transmission range is approximately 250 meters. The medium access control (MAC) layer is modeled according to IEEE 802.11a, which is widely recognized for real-world implementation. To more practically define the communication range, the RSSI threshold is set to −80 dBm. We set the value of data rate $r = 250$ kbps, the processing circuit power for transmitting $\mathtt{P}_{\mathtt{pt}} = 0.858$ mW, the transmission power required to send data from node i to the node j $\mathtt{P}_{\mathtt{tx}}^{(i,j)} = 17.33$ mW, and the efficiency of the power amplifier κ is equal to 1, respectively. The packet size x^t varies by type: INFO packets = 49 bytes, CHI packets = 55 bytes, and JOIN packets = 43 bytes. This simulation setup effectively reflects the diverse conditions encountered in practical MANET environments. Thus, we are able to conduct a thorough evaluation of the protocol, ensuring that its performance is accurately assessed and optimized.

Furthermore, to evaluate the performance of the proposed FL framework, we assess both classification and regression problems using relevant metrics. For the classification problem, we measure the test accuracy by comparing the predicted classes with the actual classes from the test set. The test accuracy is calculated as the ratio of correct predictions to the total number of predictions which is defined as follows [37]

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}. \tag{19}$$

A higher accuracy value indicates a better classification performance. For the regression problem, we measure the accuracy of the predicted optimal values by comparing them with the actual output data from the test set. This evaluation is carried out by calculating the root mean square error (RMSE),

TABLE 8. Simulation environments and network parameters.

Parameters	Value
Simulator	NS3 version 3.35
Simulation area	1000×1000 m^2
Simulation time	110 s
Mobility model	RPGM [35] & RWP [36]
Session length	5 s
Number of nodes, N = {AP + UE}	[40, 80]
Node's speed range	[15:15:75] (km/h)
Max.Transmission range	250 m
RSSI threshold	−80 dBm
MAC protocol	802.11a
Data rate, r	250 kbps [28]
Processing circuit power, $\mathtt{P}_{\mathtt{pt}}$	0.858 mW [28]
Transmission power, $\mathtt{P}_{\mathtt{tx}}^{(i,j)}$	17,33 mW [28]
Power amplifier efficiency, κ	1 [28]
Packet size, x^t	INFO: 49 bits, CHI: 55 bits, JOIN: 43 bits

which is used as a metric for accuracy in our framework. The RMSE can be written as [38]

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(\zeta_{(j)} - \widetilde{\zeta}_{(j)})^2}, \tag{20}$$

where n represents the number of samples in the test set, $\zeta_{(j)}$ represents the the predicted optimal value of the j-th observation in the dataset, and $\widetilde{\zeta}_{(j)}$ represents the actual value for the j-th observation in the dataset, respectively. A smaller RMSE indicates a closer match between the predictions and the observations. In addition, performance improvement is evaluated over multiple sessions using the FL approach. Each number of sessions involves aggregating model updates from multiple nodes to improve the global model iteratively. The number of sessions is a crucial factor in the convergence and performance of the FL model.

B. PERFORMANCE METRICS

To evaluate the performance of the proposed CFLC protocol, the following metrics are considered [28], [39], [40]:

- The number of cluster heads: it measures the average number of cluster heads in each session and the total number of average clusters obtained in the network.
- The number of cluster head changes: it measures the average number of cluster heads that change per cluster per session.
- The number of intermediate nodes: it measures the average number of intermediate nodes, including the number of cluster heads and gateways in each session.
- The number of broken links between intermediate nodes: it measures the average number of broken links between gateways and their cluster heads in each session. It serves as an indicator of the stability of the connections between gateways and their corresponding cluster heads.
- Control overhead: it measures the average number of control packets used for the clustering process in each session and each node.

- The average energy consumption: it measures the average dissipated energy of a node per session during the clustering process.
- The average execution time: it measures the mean time required to complete the local model training and subsequently aggregate these models.

These metrics collectively offer a thorough evaluation of the proposed CFLC protocol, covering critical aspects of IoT-enabled MANETs with CF-mMIMO operation, including quality, efficiency, and network stability. To demonstrate the advantages of the proposed CFLC protocol, we compare its performance against two established protocols which are the lowest ID (LI) clustering proposed by [16] and the highest connectivity (HC) clustering proposed by [17] under identical environmental conditions and parameters. Additionally, we compare the RPGM and RWP mobility models to evaluate their performance across different scenarios. By analyzing the CFLC protocol's performance across these metrics, researchers can better assess its effectiveness for real-world applications and make informed decisions about its deployment and optimization.

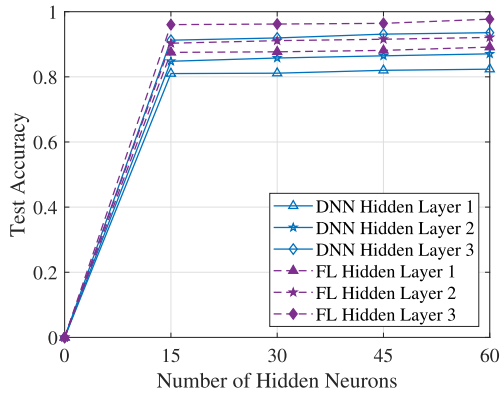
C. NUMERICAL RESULTS

In this subsection, we present the simulation results of the proposed CFLC protocol. We consider both classification and regression problems in our evaluation of the FL framework, utilizing both traditional DNN and FL approaches. For the classification problem, we generate a dataset of 10K samples and 100K samples for the regression problem, utilizing 90% for training and 10% for validation. Additionally, we produce 100 distinct datasets to evaluate the performance of our trained models. The objective is to assess the ability of the models to predict the optimal value when presented with previously unseen data. The DNN model is trained on the centralized dataset while the FL approach trains a global model by aggregating updates from multiple nodes. We employed the Adam optimizer, a widely used choice for training DNN models due to its adaptive learning rate and effective handling of sparse gradients. For classification, we used categorical crossentropy as the loss function to effectively measure the difference between true and predicted probabilities in multi-class tasks. For regression, mean squared error (MSE) is applied to calculate the average squared difference between predicted and actual values, suitable for continuous numerical predictions [41]. The detailed parameters that we used to examine DNN for both classification and regression problems are shown in Table 9.

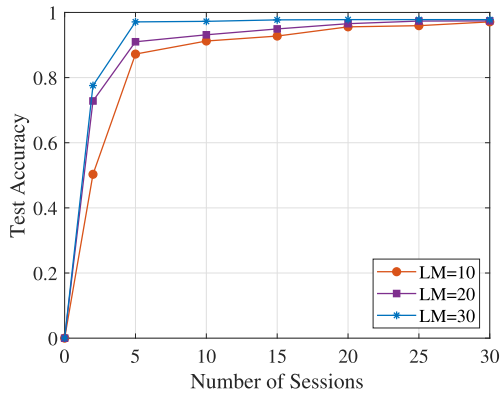
Fig. 8 shows the test accuracy performance of the classification of CH and CM in DNN and FL. In Fig. 8(a), we analyze the impact of the number of hidden neurons on the DNN and FL with the different numbers of hidden layers. As shown in Fig. 8(a), the test accuracy for both DNN and FL improves significantly with the increase of the number of hidden layers, especially when the number of hidden neurons increases from 0 to 60. This trend highlights the enhanced performance achieved by incorporating more hidden neurons and layers into the model. In particular, the accuracy of FL models was better than that of DNN models because updates were aggregated from several nodes, each having potentially unique and diverse samples. Such aggregation results in a more generalized model that performs well across different data distributions. In contrast, DNNs trained on a single centralized dataset may not capture the diversity present in the decentralized data, leading to comparatively lower performance. Fig. 8(b) shows the test accuracy of the FL global model with varying numbers of sessions and local models. The test accuracy increases significantly during the initial sessions and then gradually converges to a high value

TABLE 9. DNN training parameters.

Parameters	Classification	Regression
Dataset Size	10,000	100,000
Batch Size	256	256
Epoch	50	50
Optimizer	Adam	Adam
Learning Rate	0.00001	0.00001
Loss Function	Categorical Crossentropy	Mean Squared Error



(a) Test accuracy DNN vs FL with varying numbers of hidden neurons and layers



(b) Test accuracy of FL global model with varying number of sessions and local models (LM)

FIGURE 8. Comparison of test accuracy performance in DNN and FL for the classification of CH and CM.

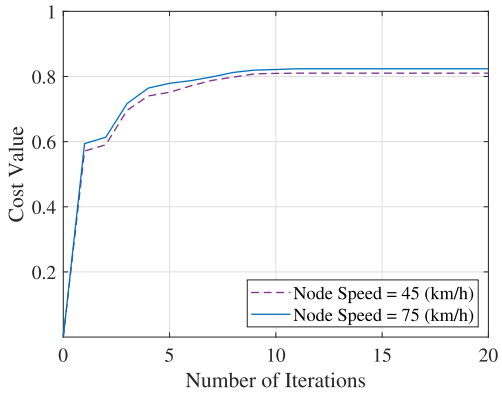
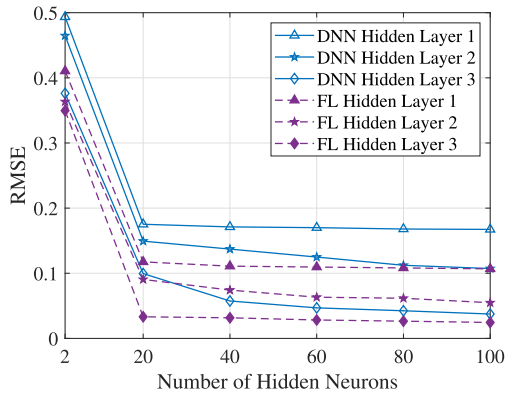
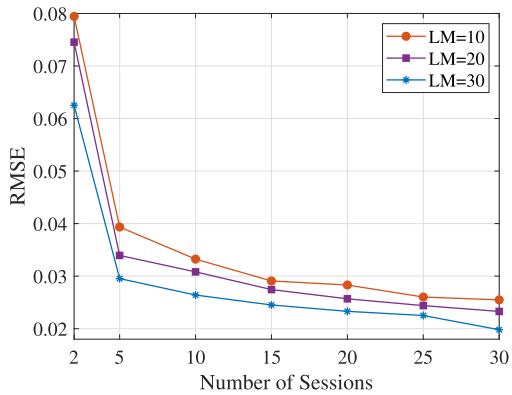


FIGURE 9. The convergence using a conventional algorithm for clustering cost function value maximization.



(a) RMSE DNN vs FL with varying numbers of hidden neurons and layers



(b) RMSE of FL global model with varying number of sessions and local models (LM)

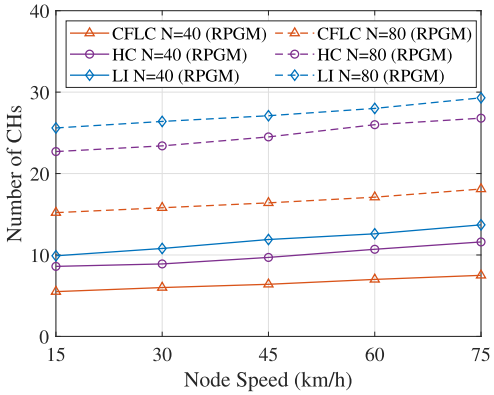
FIGURE 10. Comparison of RMSE performance in DNN and FL for cluster formation optimization.

close to one. This demonstrates that the performance of the FL model improves and stabilizes as the number of sessions increases. In addition, higher local models achieve higher accuracy than fewer local models because they bring in more varied data, helping the model learn better and make more accurate predictions. Despite this, all configurations achieve high accuracy, indicating that the FL global model effectively leverages the data from all nodes to achieve reasonable performance.

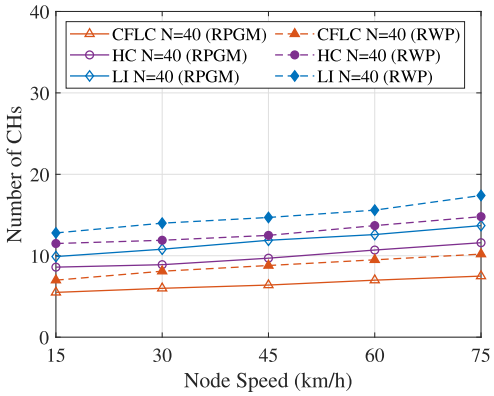
To analyze the optimization problem for cluster formation, we evaluate the impact of the number of iterations on the maximum cost using the conventional algorithm. Fig. 9 demonstrates the convergence of the cost function value for the cluster formation problem in Eq. (1), showing that the conventional algorithm reaches an optimal value within the 11th iteration. Notably, as the node speed increases, the cost value also increases, indicating consistent convergence within the 11th iteration despite higher speeds. This proves the ability of the algorithm to handle an increase in speed without affecting its convergence time.

Fig. 10 shows the comparison of RMSE performance in DNN and FL for cluster formation optimization. In Fig. 10(a), we examine the effect of varying the number of hidden neurons on the DNN and FL with the different numbers of hidden layers. The RSME for both DNN and FL decreases significantly with the increase in the number of hidden layers, especially when the number of hidden neurons increases from 2 to 100. It is evident that models with more hidden layers show lower RMSE values, indicating better performance. Notably, the FL models consistently outperform the DNN models across all configurations. This superior performance of FL models can be attributed to the aggregation of model updates from multiple nodes, which allows the FL model to capture diverse data and thus generalize better. Fig. 10(b) shows the RMSE of the FL global model with varying numbers of sessions and local models. The graph indicates a rapid decrease in RMSE during the initial sessions, followed by a gradual convergence to a low RMSE. This trend demonstrates that the performance of the FL model improves significantly with the increase in the number of sessions as the model benefits from the continuous aggregation of updates from multiple nodes. In addition, higher local models achieve lower RMSE than those with fewer local models. This indicates that increasing the number of local models improves model accuracy by reducing prediction errors more effectively. However, all configurations achieve low RMSE, indicating that the FL model effectively leverages the data from all nodes for cluster formation optimization.

Fig. 11 shows the number of CHs in the proposed CFLC protocol compared to two well-known protocols as a function of node speed under varying conditions. In Fig. 11(a), we analyze the number of CHs required for different numbers of nodes N = 40 and N = 80 with the same mobility model (RPGM). As the number of nodes increases, the number of CHs required by all protocols increases. This increase is more noticeable at a higher speed, where frequent changes in network topology necessitate more CHs to maintain effective clustering. The proposed CFLC protocol demonstrates superior efficiency, requiring fewer CHs even as the network size doubles and node speeds increase.

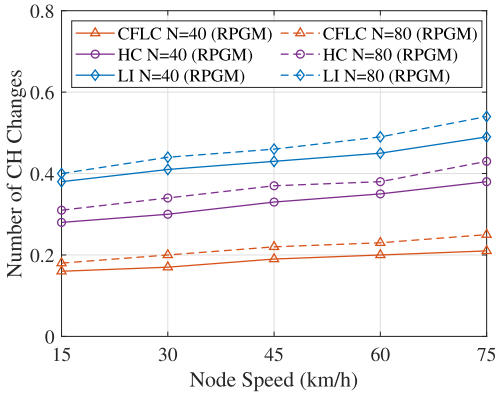


(a) Number of CHs for different numbers of nodes with the same mobility

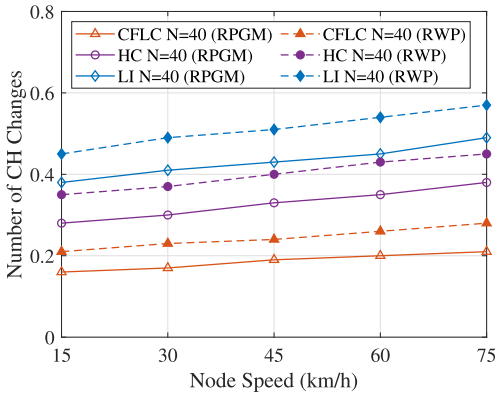


(b) Number of CHs for the same number of nodes with different mobility

FIGURE 11. The number of CHs versus node speed under varying conditions.



(a) Number of CH changes for different numbers of nodes with the same mobility

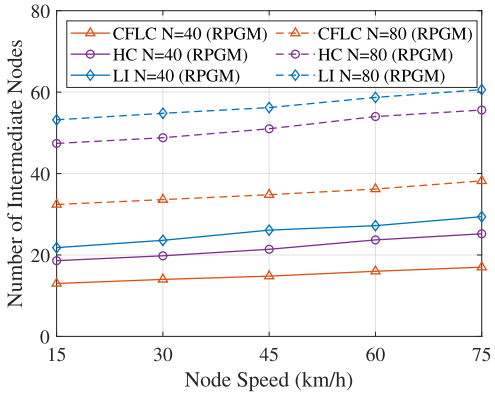


(b) Number of CH changes for the same number of nodes with different mobility

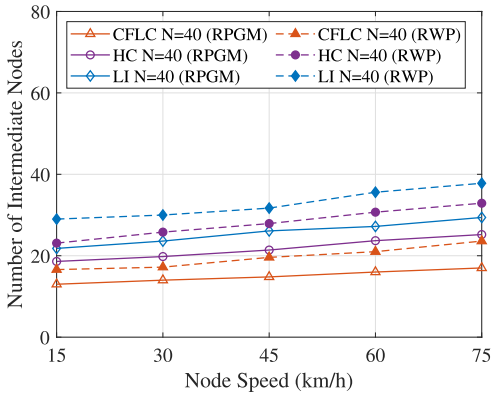
FIGURE 12. The number of CH changes versus node speed under varying conditions.

In contrast, both the HC and LI protocols show a more significant increase in CHs with the LI protocol requiring substantially more CHs, especially at higher node speeds. The superior performance of CFLC can be attributed to its use of FL to optimize cluster formation. In Fig. 11(b), we analyze the number of CHs required for the same number of nodes N = 40 with different mobility models (RPGM and RWP). The results show that the RWP model generally requires more CHs for all protocols compared to the RPGM model. This is expected, as the RWP model results in more unpredictable node movements, leading to more frequent changes in cluster topology and requiring more CHs to maintain network stability. Meanwhile, in the RPGM model, the nodes move cohesively in groups, which means they tend to follow similar trajectories and move together as a unit. However, the CFLC protocol still maintains the lowest number of CHs. This demonstrates the proposed CFLC protocol's efficiency in managing dynamic networks with varying mobility patterns, proving to be highly effective in enhancing both stability and scalability.

Fig. 12 shows the number of CH changes as a function of node speed under varying conditions. In Fig. 12(a), we analyze the number of CH changes for different numbers of nodes with the same mobility model. As node speed increases from 15 km/h to 75 km/h, all protocols exhibit an increase in the number of CH changes due to nodes moving faster and changing positions more frequently, leading to more frequent CH changes. However, the CFLC protocol exhibits a significantly lower rate of increase in CH changes, demonstrating its robustness in handling high-mobility scenarios even when the number of nodes N increases from 40 to 80. The HC and LI protocols, particularly the LI protocol, exhibit a much higher number of CH changes at higher node speeds, reflecting their reduced stability in dynamic environments. In Fig. 12(b), the number of CH changes is analyzed for the same number of nodes with different mobility models. The results indicate that the number of CH changes is generally lower for all protocols under the RPGM model compared to the RWP model. The reason is that the RPGM model involves nodes moving cohesively in groups, creating a more predictable and stable network topology that reduces the need for frequent CH changes. Consequently, the CFLC protocol demonstrates the lowest number of CH changes when using the RPGM model outperforming both the HC and LI protocols, highlighting its efficiency and robustness in dynamic networks.

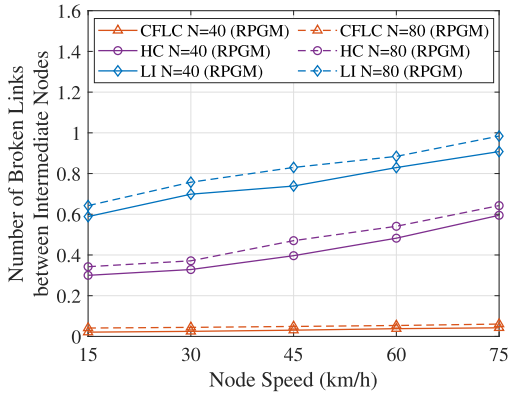


(a) Number of intermediate nodes for different numbers of nodes with the same mobility

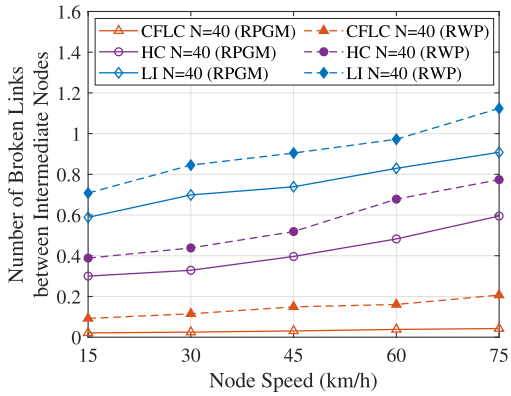


(b) Number of intermediate nodes for the same number of nodes with different mobility

FIGURE 13. The number of intermediate nodes versus node speed under varying conditions.



(a) The number of broken links between intermediate nodes for different numbers of nodes with the same mobility

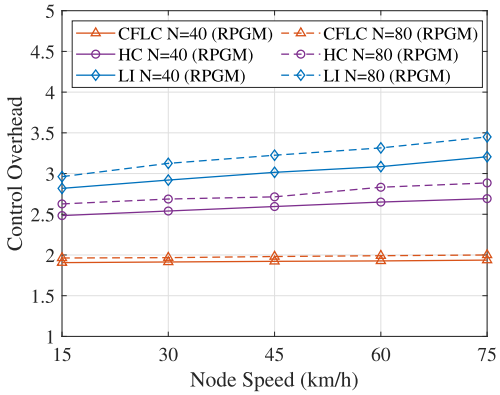


(b) The number of broken links between intermediate nodes for the same number of nodes with different mobility

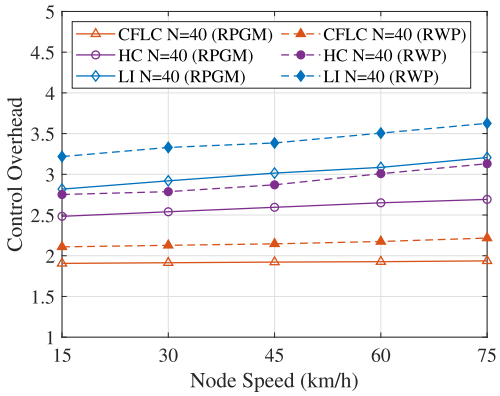
FIGURE 14. The number of broken links between intermediate nodes versus node speed under varying conditions.

Fig. 13 shows the number of intermediate nodes as a function of node speed under varying conditions. The number of intermediate nodes for different numbers of nodes with the same mobility model is analyzed in Fig. 13(a). In the proposed CFLC protocol, this number implies the total number of CHs and GWs that range from 13 to 17 for N = 40. Given that there are approximately 6 CHs as shown in Fig. 11(a), the number of GWs is around 8, indicating that each cluster has about 1.33 GWs on average. This relatively low number of GWs per cluster indicates that the CFLC protocol efficiently maintains connectivity between clusters. Overall the number of intermediate nodes increases as the number of nodes increases. The network with N = 80 consistently requires more intermediate nodes than the network with N = 40 across all speeds. The reason is that a larger network typically needs more relay nodes to facilitate communication between distant nodes and maintain connectivity. Additionally, as node speed increases, the number of intermediate nodes required by all protocols slightly increases due to more frequent changes in the network topology, which demands more relay nodes to sustain stable routes. Furthermore, we analyze the number of intermediate nodes for the same number of nodes but with different mobility as depicted in Fig. 13(b). The results show that the number of intermediate nodes is generally higher under the RWP model than the RPGM model for all protocols. The reason is that the RWP model involves more random and independent node movements, leading to frequent disruptions in direct communication paths. Despite this, the CFLC protocol consistently maintains the lowest number of intermediate nodes compared to the HC and LI protocols, demonstrating its effectiveness in achieving optimized cluster formation.

Fig. 14 depicts the number of broken links between intermediate nodes versus node speed under varying conditions. In Fig. 14(a), we analyze the number of broken links between intermediate nodes for different numbers of nodes with the same mobility model. The results show that as the number of nodes increases, the number of broken links between intermediate nodes also increases for all protocols. This trend is more noticeable at higher node speeds, where the network topology becomes more dynamic, resulting in a greater likelihood of link breakages. The LI protocol exhibits the highest number of broken links, followed by the HC protocol



(a) Control overhead for different numbers of nodes with the same mobility



(b) Control overhead for the same number of nodes with different mobility

FIGURE 15. Control overhead versus node speed under varying conditions.

while the proposed CFLC protocol consistently shows the lowest number of broken links. Furthermore, the number of broken links between intermediate nodes for the same number of nodes with different mobility is examined in Fig. 14(b). As shown in Fig. 14(b), the number of broken links under the RPGM model for all protocols also shows lower results than the RWP model. This is due to the higher stability of the RPGM model than the RWP model. Despite this, the proposed CFLC protocol continues to maintain the lowest number of broken links under both mobility models. The reason is that factors such as mobility similarity and link quality help to optimize cluster formation in dynamic network conditions.

Fig. 15 presents the control overhead versus node speed under varying conditions. In Fig. 15(a), we examined the control overhead for different numbers of nodes with the same mobility. The results show that as the number of nodes increases, the control overhead increases for all protocols. The reason is that a larger network size requires more frequent updates and coordination among nodes to maintain a stable cluster. The increase in control overhead is more evident at higher node speeds, where the dynamic changes in topology demand additional control packets to ensure

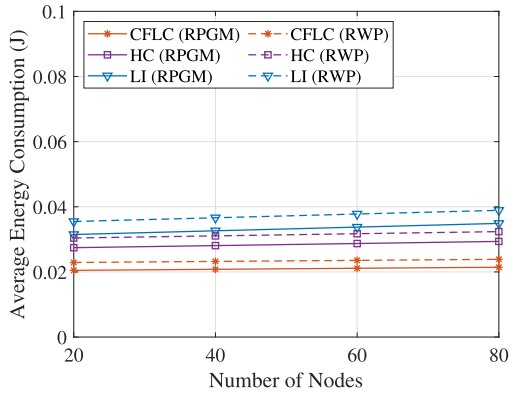


FIGURE 16. The average energy consumption versus number of nodes.

network stability. Among the protocols, the LI protocol exhibits the highest control overhead, followed by the HC protocol while the proposed CFLC protocol maintains the lowest control overhead across all speeds and network sizes. Moreover, we also analyze the control overhead for the same number of nodes under different mobility models as shown in Fig. 15(b). The results further confirm that the RPGM model consistently exhibits lower control overhead than the RWP model. This reason is that the RPGM model features more predictable and cohesive movement patterns, resulting in fewer topology changes, thereby reducing the need for frequent control packet exchanges. This proves that the RPGM model is more efficient than the RWP model, providing better overall network performance. Despite these differences, the proposed CFLC protocol consistently achieves the lowest control overhead under both mobility models, underscoring its efficiency in minimizing control packet exchanges and maintaining optimal network performance. This efficiency is particularly evident in larger networks and at higher node speeds.

Fig. 16 illustrates the average energy consumption as a function of the number of nodes with a node speed of 45 km/h. As the number of nodes increases from 20 to 80, the average energy consumption for all protocols also increases. This is due to the higher number of transmissions and receptions required as the network density grows, which leads to higher average energy consumption. However, the proposed CFLC protocol demonstrates consistently lower average energy consumption compared to the HC and LI protocols under both mobility models, even as the number of nodes increases. The reason is that FL helps to predict optimal CH selection and forms stable clusters by analyzing node behavior such as speed and remaining energy. These optimizations reduce the frequency of CH re-elections and re-clustering, leading to lower control overhead. In contrast, the LI protocol shows the highest average energy consumption due to an inefficient and less optimized clustering process, which leads to more frequent re-clustering and increased control overhead. Similarly, while the HC protocol performs better than LI, it relies on basic connectivity criteria for CH selection. This can lead to less optimal CH selection in

TABLE 10. Average execution time of conventional vs FL approach.

Number of Nodes	Classification		Regression	
	Conven (sec)	FL (sec)	Conven (sec)	FL (sec)
20	8.95	0.727	25.15	0.625
40	17.18	0.728	33.53	0.637
60	25.43	0.742	40.77	0.651
80	33.83	0.752	45.64	0.783

dynamic environments, resulting in higher average energy consumption. The advantages of the proposed CFLC protocol in terms of average energy consumption are particularly evident under the RPGM model, where the model's stability enables it to outperform the conventional HC and LI protocols.

Finally, Table 10 compares the execution time for the conventional versus the proposed FL approach for different numbers of nodes with both the classification and regression problem. The results demonstrate the significant efficiency of the FL approach in comparison to the conventional one, especially when the number of nodes increases. The FL approach consistently maintains an execution time of less than one second even when the node increases to 80. In contrast, the conventional approach's execution time grows with the increased number of nodes. For example, in the classification problem with 80 nodes, the conventional approach takes 33.83 seconds while the FL approach completes the same task in just 0.752 seconds. Similarly, in the regression task, the conventional approach requires 45.64 seconds for 80 nodes while the FL approach achieves the result in only 0.783 seconds. The reduction in execution time with the FL approach can be attributed to the DNN model, which can efficiently handle large datasets. The multiple layers and neurons within the DNN model smoothen computations, allowing complex modeling. Once the model is trained, predictions are less computationally intensive, resulting in faster execution times. This makes the FL approach in the proposed CFLC protocol particularly suitable for implementations, including URLLC services, where real-time decision-making is critical as the network size increases. The significant reductions in average execution time highlight the practicality of the protocol, ensuring it can be effectively deployed in dynamic, large-scale, and high-mobility networks.

The results above show that the proposed CFLC protocol consistently outperforms the HC and LI protocols across various performance metrics, especially using the RPGM model. The proposed CFLC protocol performs well under varying conditions, including different network sizes, node speeds, and mobility models. In detail, the proposed FL is effectively utilized to address the classification of nodes as CHs and CMs and optimize the cost function weights for cluster formation. Consequently, the proposed CFLC protocol enhances network stability and minimizes the need for frequent re-clustering, making it highly efficient in dynamic and large-scale network environments.

V. CONCLUSION

In this paper, we proposed a novel CFLC protocol to support multicast routing in IoT-enabled MANETs with CF-mMIMO. The proposed CFLC protocol leveraged cross-layer and FL to enhance network stability and connectivity by optimizing CH selection and cluster formation. The cross-layer design integrated physical layer information such as mobility (speed and direction), position, channel capacity, and remaining energy with network layer information (i.e., connectivity) to maximize the cost function value for cluster formation. Furthermore, the FL model was employed to improve the clustering performance and satisfy future mobile network requirements. In detail, during the CH selection step, the FL decided which nodes should be elected as CHs or CMs by using classification. During the cluster formation step, the FL addressed a regression problem by optimizing the cost function weights for parameters such as mobility similarity, link quality, remaining energy, and channel capacity to decide which CH each node should follow. The simulation results showed that the proposed CFLC protocol outperformed the benchmark protocols in terms of connectivity, scalability, and control overhead. Additionally, the results indicated that the CFLC protocol performs particularly well when using the RPGM model, highlighting its advantage over the RWP mobility model in maintaining network stability and connectivity. For future work, we aim to extend this approach to develop a multicast routing protocol, which is essential for IoT-enabled MANETs with CF-mMIMO environments. Additionally, we will investigate the impact of the bandwidth on distributed learning models by employing a cross-layer design to optimize performance across different layers. Finally, real-world testing will be conducted to gain practical insights and validate the proposed CFLC protocol, thereby complementing our simulation results.

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AMALIA AMALIA received the B.S. degree in informatics engineering and the M.S. degree in electrical engineering from Politeknik Negeri Malang, Malang, Indonesia, in 2016 and 2018, respectively. She is currently pursuing the Ph.D. degree with the Department of Software and Communications Engineering, Graduate School, Hongik University, Sejong, Republic of Korea. Her current research activities are focused on mobile wireless networks, clustering, QoS routing, QoS multicast routing, and machine learning applications.

YUSHINTIA PRAMITARINI (Graduate Student Member, IEEE) received the B.S. degree in telecommunications engineering from the Electronic Engineering Polytechnic Institute of Surabaya, Surabaya, Indonesia, in 2011, and the M.S. degree in electrical engineering from the Institut Teknologi Sepuluh Nopember (ITS), Surabaya, in 2013. She is currently pursuing the Ph.D. degree with the Department of Software and Communications Engineering, Graduate School, Hongik University, Sejong, Republic of Korea. Her current research activity is focused on mobile wireless networks, clustering, QoS routing, QoS multicast routing, machine learning applications, and physical layer security (PLS). In 2024, she received the IEEE International Conference on Artificial Intelligence in Information and Communication Best Paper Award.

RIDHO HENDRA YOGA PERDANA (Graduate Student Member, IEEE) received the B.S. degree in telecommunications engineering from the Electronic Engineering Polytechnic Institute of Surabaya, Surabaya, Indonesia, in 2012, and the M.S. degree in electrical engineering from the Institut Teknologi Sepuluh Nopember (ITS), Surabaya, in 2014. He is currently pursuing the Ph.D. degree with the Department of Software and Communications Engineering, Graduate School, Hongik University, Sejong, South Korea. His main research focuses on the mathematical modeling of beyond 5G networks and optimization/machine learning for wireless communications. He received the Best Paper Award of the IEEE International Conference on Artificial Intelligence in Information and Communication, in 2024.

KYUSUNG SHIM (Member, IEEE) received the B.S. degree in computer and information communications engineering, the M.S. degree in information system, and the Ph.D. degree in electronics and computer engineering from Hongik University, Sejong, South Korea, in 2012, 2017, and 2021, respectively. After graduation, he joined the Faculty of the School of Computer Engineering and Applied Mathematics, Hankyong National University, Anseong, South Korea, where he is currently an Assistant Professor. His main research interests include AI-based wireless communications and AI-based multicast routing protocol.

BEONGKU AN (Senior Member, IEEE) received the B.S. degree in electronic engineering from Kyungpook National University, Republic of Korea, in 1988, the M.S. degree in electrical engineering from Polytechnic University (NYU), NY, USA, in 1996, and the Ph.D. degree in electrical engineering from New Jersey Institute of Technology (NJIT), NJ, USA, in 2002. After graduation, he joined the Department of Software and Communications Engineering, Hongik University, Republic of Korea, where he is currently a Professor. From 1989 to 1993, he was a Senior Researcher with RIST, Republic of Korea. His current research interests include mobile wireless networks and communications, such as ad-hoc and sensor networks, cognitive radio networks, cellular networks, and the IoT. In particular, he is interested in cooperative transmission, QoS routing, QoS multicast routing, energy harvesting, physical layer security (PLS), visible light communication (VLC), cross-layer technology, 5G/beyond 5G, machine learning/deep learning applications. He was listed in Marquis Who's Who in Science and Engineering and Marquis Who's Who in the World. He was a president of IEIE Computer Society, in 2012 and the General Chair of the international conference ICGHIT, from 2013 to 2017.